

TATIANA KULAGINA

London, United Kingdom

kulaginao.tatiana@gmail.com ◇ github.com/bilunovat ◇ linkedin.com/in/tkulagina

Backend engineer with an **MSc in Computer Science**, specialising in **Java and Spring Boot**. Started with **C** to build a strong foundation. Built an automated least-privilege policy framework as my dissertation.

TECHNICAL SKILLS

Languages: Java, C, SQL, MySQL, HTML/CSS, JavaScript

Backend: Spring Boot, Spring AOP, REST API, RabbitMQ, Maven, JUnit

Infrastructure & Observability: Docker, Prometheus, Grafana, OPA, GitHub Actions, Git

EDUCATION

Newcastle University, Newcastle Upon Tyne

September 2025 - September 2026

Computer Science, MSc. Current average: 81% (distinction track)

Vice Chancellor's Global Scholarship 2025

Moscow State Linguistic University, Moscow

September 2019 - July 2023

Translation and Interpretation, English and Portuguese, BA

COURSES AND PERSONAL PROJECTS

Dissertation Project | Java

June - August, 2026

Automated Least-Privilege Policy Synthesis for Containerised Microservices

- Designed a Traffic Observer using Spring AOP to capture inter-service communication, adding only 1.5–5.0 ms overhead per request.
- Implemented an automated Policy Inference Engine that translates observed traffic into OPA rules.
- Built a Policy Drift Detector to validate live traffic against the inferred rules and alert on anomalies in 102 ms. Flagged zero false positives across 151 legitimate interactions.
- Orchestrated a Docker Compose microservice environment to simulate traffic scenarios.

Coffee Web App | Java, Spring Boot

April, 2026

- Co-developed a full-stack web application that allows users to place an order online and staff members to manage order processing status.
- Solely owned the order processing engine and full backend integration testing with 87% coverage.
- Containerised app with Docker and enforced code style with a GitHub Actions CI pipeline.

Split Keyboard Project

August, 2025; January, 2026

Designed and built a custom split keyboard from 3D modelling and soldering to programming. Developed firmware in CircuitPython.

21 School (42 Network), Moscow | C

April - September 2025

Intensive peer-to-peer coding school

Project SimpleBash: Engineered custom implementations of cat and grep CLI utilities with flag parsing and combination logic. Strengthened file I/O, dynamic memory management, and low-level programming.

Project 21string+: Led a team of four to successfully implement the functionality of string.h, sscanf and sprintf. Personally developed most of the sscanf functionality with tests achieving 98% code coverage.